Code No: 156CA **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
NETWORK PROGRAMMING
(Common to CSE, IT)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

JNTUH Used papers 2023

PART – A

(25 Marks)

1.a)	Define concurrent server.	[2]
b)	Write a program to determine the host bye order.	[3]
c)	Write a function of shutdown of a server host.	[2]
d)	What is the need of I/O model?	[3]
e)	What is the use of Ancillary Data?	[2]
f)	What is the syntax of setsockopt( )?	[3]
g)	What is uname( ) function?	[2]
h)	Differentiate between unicast and multicast addressing.	[3]
i)	What is the significance of ping program?	[2]
j)	How are pseudo terminals different from control terminals?	[3]

PART – B

(50 Marks)

2. Describe the any 5 elementary TCP socket functions. [10]

OR

3.a) Describe the four way handshaking mechanism in TCP connection establishment.
b) Explain about the standard internet services. [5+5]

4. Differentiate between synchronous and asynchronous I/O models. [10]

OR

5. Explain about the shutdown function and poll function. [10]

6. Explain IPV6 and ICMPV6 socket options in detail. [10]

OR

7.a) What are the different socket states in a network communication?
b) How to modify the socket's behavior in response to error conditions? [5+5]

8.a) Write a program that creates a daemon process using the daemon_init() function, which logs messages using the syslog() function and the syslogd daemon.
b) Explain about the Multicast Socket Options. [5+5]

OR

9.a) How does the Simple Network Time Protocol (SNTP) is used in multicasting for time synchronization of a network?

b) What is a daemon process and how does it differ from a normal process in a Unix operating system? What mechanism does DNS employ to efficiently return the IP address for the given domain name. [5+5]

10.a) What are terminal modes and control terminals, and how can they be set an manipulated in a program to control the behavior of a terminal and interact with processes running in the foreground and background of the program.

b) What is the rlogin program and how does it provide remote login capabilities, allowing a user to log into a remote machine and interact with its shell as if they were directly connected to the machine. [5+5]

OR

11.a) What are the benefits of using a packet capture library when working with network packets, and how does it provide an easy and efficient way to analyze and manipulate network traffic?

b) What is the purpose of the traceroute program and how does it work to determine the route taken by packets from a source to destination in a network. [5+5]

---ooOoo---

JNTUH Used papers 2023